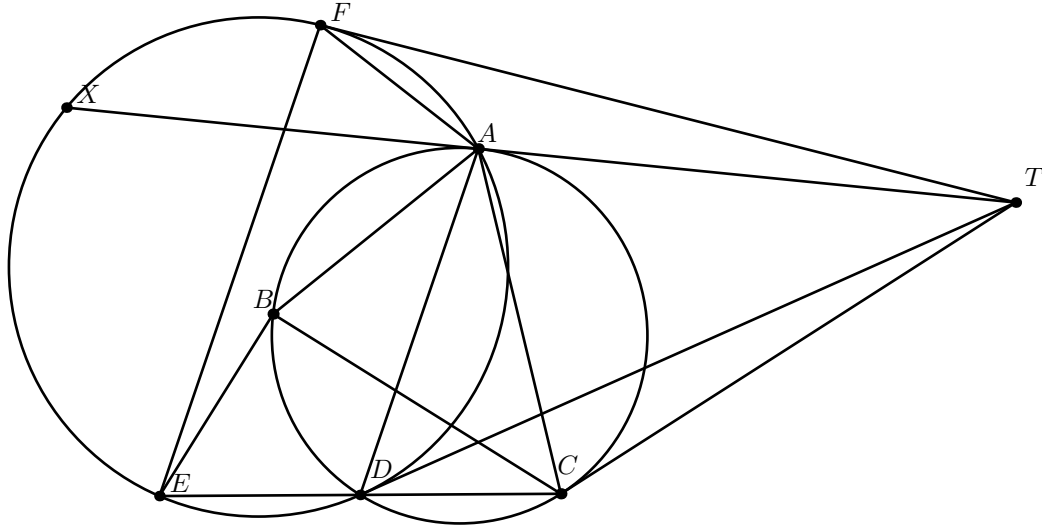


NMTC Problem-Set 2: Solutions

Instructions: The following are the solutions for Problem Set 2. The questions haven't been included this time so it's recommended to read this document along with the problem set document. The sources to the problems can be found on the last page of the document.

Refer to the given diagram for the solution:



Proof. (a) For ease, let $\angle BAC = 2\alpha$, $\angle ABC = 2\beta$, $\angle ACB = 2\gamma$. Then from yesterday's solution(or quickly derivable from the fact that equal chords subtend equal angles), we have that $\angle DBC = \angle DCB = \alpha$, and hence $DC = BC$.

(b) Since $AD \parallel EF$, and $FADE$ is a cyclic quadrilateral, we have the following:

It follows that $AF = DE$, and we are done.

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We have:

$$\begin{aligned}\angle DCT &= \angle DCB + \angle BCA + \angle ACT \\ &= \alpha + 2\gamma + 2\beta\end{aligned}$$

Note that by part (b), $\angle FAD = \angle ADE = 180^\circ - \angle ADC = 180^\circ - 2\beta$.
We then have:

$$\begin{aligned}\angle FAT &= 180^\circ - \angle FAX \\ &= 180^\circ - (\angle FAD - \angle BAD - \angle XAB) \\ &= 180^\circ - ((180^\circ - 2\beta) - (\alpha)) - (2\gamma) \\ &= \alpha + 2\gamma + 2\beta\end{aligned}$$

where we applied Tangent-Chord theorem to find $\angle XAB$. Hence $\angle FAT = \angle DCT$, as needed.

- (d) We first show that $\triangle TFD \sim \triangle TAC$. Observe that by part (c), $TF = TD$, and $TA = TC$, so $\frac{TF}{TA} = \frac{TD}{TC}$. Also,

$$\begin{aligned}\angle FTD &= \angle ATD + \angle FTA \\ &= \angle ATD + \angle DTC \\ &= \angle ATC\end{aligned}$$

Hence, By *SAS* similarity criteria, we're done. Next. Observe that:

$$\begin{aligned}\angle FED &= 180^\circ - \angle FAD = 2\beta \\ &= \angle ABC \\ &= \angle CAT \\ &= \angle DFT\end{aligned}$$

Hence, by the converse of the Tangent-Chord theorem applied to $\angle DFT$ with $\triangle EFD$, we have that FT is tangent to ω , as needed.

□

Problem 2: Counting

Proof. 1. We see that $N = \sum_{i=0}^{11} \binom{11}{i} \binom{12}{i+1}$. Now, since

$$\binom{12}{i+1} = \binom{12}{12-(i+1)} = \binom{12}{11-i}$$

We get $N = \sum_{i=0}^{11} \binom{11}{i} \binom{12}{11-i} = \binom{23}{11}$. Hence:

$$N = \frac{23 \times 22 \times 21 \times 20 \times 19 \times 18 \times 17 \times 16 \times 15 \times 14 \times 13 \times 12}{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

Disregarding the primes in the numerator and simplifying gives

$$N = 23 \times 19 \times 17 \times 14 \times 13$$

so the sum of the prime factors of N is

$$23 + 19 + 17 + 13 + 7 + 2 = 81$$

2. (a) Suppose that $2^a, 2^b, 2^c$ are our 3 terms in a geometric progression. Then,

$$\frac{2^b}{2^a} = \frac{2^c}{2^b} \iff 2^{2b} = 2^{a+c} \iff 2b = a+c \iff b = \frac{a+c}{2}$$

Thus, every choice of an unordered pair (a, c) , where a and c have the same parity, corresponds to a unique geometric progression (Why?). The restriction that a, c have the same parity means we must have $(a, c) \in \{1, 3, 5, \dots, 2019\}$ or $(a, c) \in \{2, 4, 6, \dots, 2020\}$. Each of these sets has 1000 elements, so the total number of choices is $2 \binom{1000}{2} = 999000$.

- (b) For the challenge part, you can see from the previous solution that the answer doesn't depend on n , only on k . We split into 2 cases, one where k is even, and the other where k is odd.
- i. If k is even, we write $k = 2m$, for some $m \in \mathbb{N}$. By the previous solution, each unordered pair (a, c) , where a, c have the same parity, determines a geometric progression. Thus, $(a, c) \in \{1, 3, \dots, 2m-1\}$ or $(a, c) \in \{2, 4, \dots, 2m\}$, and each of these sets has m elements. This gives a total of $2\binom{m}{2}$ possibilities. In terms of k , this is $\frac{k(k-2)}{4}$.
 - ii. In the case that k is odd, we write $k = 2m - 1$ for some, $m \in \mathbb{N}$. Similar to the previous part, we have $(a, c) \in \{1, 3, \dots, 2m-1\}$ or $(a, c) \in \{2, 4, \dots, 2m-2\}$. The first set has m elements, while the second set has $m-1$ elements. This gives a total of $\binom{m}{2} + \binom{m-1}{2}$ possibilities. Expanding and writing in terms of k gives us $(\frac{k-1}{2})^2$ possibilities.

□

Problem 3: Some Divisibility

1. *Proof.* We proceed by contradiction. Assume that $a + b = p$ for some prime p . Furthermore, Let $(a^2 - 1)(b^2 - 1) = x^2$ for some integral x . We have:

$$\begin{aligned} (a+1)(a-1)(b+1)(b-1) &= x^2 \\ \iff (ab+a+b+1)(ab-a-b+1) &= x^2 \\ \iff (ab+1)^2 - p^2 &= x^2 \\ \iff (ab+1+x)(ab+1-x) &= p^2 \end{aligned}$$

So, p^2 has two distinct factors. This implies that $ab+1-x = 1$. Hence, $ab = x$. It also implies that $ab+1+x = (a+b)^2$. Hence, $2ab = p^2 - 1$. We now have:

$$\begin{aligned} a^2 + b^2 &= (a+b)^2 - 2ab \\ &= p^2 - (p^2 - 1) \\ &= 1 \end{aligned}$$

But this is absurd. Hence, we conclude that $a + b$ can never be prime, as needed. $\square$

2. *Proof.* Let $a - b = p$ for some prime p and let $(a^2 - 1)(b^2 - 1) = x^2$ for some integral x . We have:

$$\begin{aligned} (a+1)(a-1)(b+1)(b-1) &= x^2 \\ \iff (ab+a-b-1)(ab-a+b-1) &= x^2 \\ \iff (ab-1)^2 - p^2 &= x^2 \\ \iff (ab-1+x)(ab-1-x) &= p^2 \end{aligned}$$

Similar to the previous part, we get $ab-1+x = p^2$ and $ab-1-x = 1$.

Adding these equations gives $ab = \frac{p^2 + 3}{2}$. Hence:

$$\begin{aligned} 6ab - 3 &= 3p^2 + 6 \\ &= p^2 + 4 \left(\frac{p^2 + 3}{2} \right) \\ &= (a-b)^2 + 4ab \\ &= (a+b)^2 \end{aligned}$$

Which is a perfect square, as needed. $\square$

Problem 4: Some Algebra!

1. *Proof.* We Solve each part seperately.

- (a) (Adapted from the official solution) Denote the numbers in the table as follows:

a	b
c	d

We have the following relations:

$$a = bc$$

$$b = ad$$

$$c = ad$$

$$d = bc$$

Hence, $a = d$, and $b = c$. This gives us $a = b^2$, and $b = a^2$. Hence, $a = a^4$. So, $a = 0, 1$. It follows that all 2×2 *product* tables consist of all entries 0, or 1

- (b) Label the entries of the table as follows:

a	abc	c
ab	b	bc

Now, if any entry is 0, it follows that all others are 0. Hence, from now on we assume that all entries are non-zero. We get the relations:

$$\begin{aligned} a = a^2 b^2 c &\iff 1 = ab^2 c \\ b = b^3 a^2 c^2 &\iff 1 = b^2 a^2 c^2 \\ \implies ab^2 c = b^2 a^2 c^2 \\ &\iff 1 = ac \end{aligned}$$

This implies that $acb^2 = b^2 = 1$. So, $b = \pm 1$.

If $b = 1$, we get the table:

a	1	$\frac{1}{a}$
a	1	$\frac{1}{a}$

Where a is any non-zero real number.

If $b = -1$, we get the table:

a	-1	$\frac{1}{a}$
$-a$	-1	$-\frac{1}{a}$

Where again, a is any non-zero real number. These combined with the all zero table gives us all 2×3 *product* tables.

- (c) (Adapted from the official solution) Denote by a, b, c, d the elements in the table which have exactly three neighbours. We denote the remaining elements in terms of these and get the following table:

ab	a	ad
b	$abcd$	d
bc	c	cd

Once again, if any entry is zero, it follows that all others are zero. Hence, assume that all entries are non-zero.

We now have the relation:

$$a = (ab)(abcd)(ad) = a^3b^2d^2c \iff a^2b^2d^2c = 1$$

Analogously, we get $a^2b^2c^2d = 1$, $b^2c^2d^2a = 1$, and $a^2c^2d^2b = 1$. Equating the LHS of these equations gives us that $a = b = c = d$. Plugging this into any of the equations gives us $a^7 = 1 \iff a = 1$. Hence, all 3×3 product tables are all zeros and all ones tables.

□

2. *Proof.* We provide 2 proofs of this:

- (a) Only functions of the form $f(x) = x + c$ for some real c work. We can verify this easily, now we prove this. Denote by $P(x, y)$ The given functional equation. We will use this notation for convenience. Observe that $P(x, -f(x))$ gives us:

$$f(0) = 2x + f(f(-f(x)) - x)$$

The above equation tells us that $f(\text{something})$ can take all real values. In other words $f(x)$ can take all real values, meaning $f(x)$ is surjective¹.

Next, Let $f(0) = c$ observe that $P(0, y)$ gives us that:

$$\begin{aligned} f^2(y) &= f(y + f(0)) \\ &= f(y + c). \end{aligned} \tag{1}$$

¹A surjective function is a function which can take on all possible values in the set of outputs. For example, $g(x) = x - 1$ is surjective because it can achieve the value of any real number.

Now, $P(f(-c), -c)$ along with (1) gives us:

$$\begin{aligned} f(f^2(-c) - c) &= 2f(-c) + f(f^2(-c) - f^2(-c)) \\ \iff f(f(-c + c) - c) &= 2f(-c) + f(0) \\ \iff f(c - c) &= 2f(-c) + c \\ \iff 0 &= f(-c) \end{aligned} \quad (2)$$

Finally, $P(-c, y)$ along with (2) gives us:

$$\begin{aligned} f(f(-c) + y) &= -2f(0) + f(f(y) + c) \\ \iff f(y) &= -2c + f(f(y) + c) \\ \iff (f(y) + c) + c &= f(f(y) + c) \end{aligned} \quad (3)$$

Since $f(y)$ is surjective, it follows that $f(y) + c$ is surjective too. Hence, (3) is equivalent to:

$$f(x) = x + c$$

All functions of this form work and we're done.

(b) We start by letting $y = -f(x)$. This gives us

$$\begin{aligned} f(0) &= 2x + f(f(-f(x)) - x) \iff \\ f(f(-f(x)) - x) &= f(0) - 2x \end{aligned}$$

Since $f(0) - 2x$ can take on the value of any real number (simply by choosing an appropriate value of x), we see that f is a surjective function.

Now, we suppose that there are two numbers, a and b , such that $f(a) = f(b)$. We let t be a number such that $f(t) = a + b$ (Such a t exists because the function is surjective, although it may not be unique). In the original equation, we put $x = a, y = t$, and simplify to get:

$$f(f(a) + t) = 2a + f(b) \quad (1)$$

Similarly, we put $x = b, y = a + b$ to get:

$$f(f(b) + t) = 2b + f(a) \quad (2)$$

Since $f(a) = f(b)$, we get:

$$\begin{aligned} f(f(a) + t) &= f(f(b) + t) \iff \\ 2a + f(b) &= 2b + f(a) \iff \\ a &= b \end{aligned}$$

This proves that the function is injective.² Finally, in the original equation, we let $x = 0, y = y$, and this gives us:

$$f(f(0) + y) = f(f(y))$$

Since f is injective, we must have that

$$f(y) = f(0) + y \iff f(y) = c + y$$

where we defined $c = f(0)$. Thus we have proved that the only functions that can work must be of the form above. It remains to check that a function of this form actually does satisfy the above equation.

²An injective function is just one in which every number has a different output.

Plugging this function into the original equation, we have:

$$\begin{aligned}f(f(x) + y) &= 2x + f(f(y) - x) \iff \\f(c + x + y) &= 2x + f(c + y - x) \iff \\c + (c + x + y) &= 2x + (c + c + y - x) \iff \\2c + x + y &= 2c + x + y \iff \\0 &= 0\end{aligned}$$

Hence all functions of this form work, and we are done.

□

Sources:

1. 2024 Switzerland IMO TST P2 (Simplified. The original problem only told you to directly prove (d)).
2. (a) 2020 AIME I P7
(b) ARML 2001
3. 2025 PAGMO P2
4. (a) 2013 EMC Senior category P1 (Extended)
(b) 2002 Czech IMO shortlist, Problem A1